

Assignment 2 – week 2

1. Quality Criteria

Institutional authority: The source must be published by a recognized supply-chain or retail-technology vendor, an academic or industry research body, or a professional organization relevant to demand forecasting, inventory planning, or SME digitalization, such as a peer-reviewed operations-management journal, an established analyst firm, or a national or EU body supporting SME digital adoption. *A vendor's own product or marketing page does not count as a highly credible source on its own. It only qualifies if it is cited alongside, or corroborated by, an independent case study, analyst report, or academic source.*

Thematic & geographic relevance: The document must explicitly address AI-based demand forecasting, inventory optimization, or related SME technology adoption, and should be relevant to mid-sized, seasonal, variant-heavy wholesale or retail businesses, ideally within a European or Benelux context.

Timeliness: Due to the rapid evolution of AI technology and SME adoption patterns, sources must be published between 2024 and 2026.

Methodological transparency: The report must include clear citations, rely on verifiable data such as documented case studies or vendor-independent benchmarks, and avoid purely speculative or corporate-marketing narratives.

Scoring: Each source is scored 0 to 2 on each criterion above, where 0 means it fails the criterion, 1 means it partially meets it, and 2 means it fully meets it. A source is excluded outright if it scores 0 on Institutional Authority or Methodological Transparency, regardless of how it scores on the other two criteria.

2. Search Strategy

Where we searched: Standard Google Search (google.com), and Google Scholar for the second query below.

How we searched:

Query 1: "AI demand forecasting" wholesale OR retail SME case study

Query 2: AI inventory planning adoption barriers small and medium enterprises

We expect the second query, run on Google Scholar rather than standard Google, to surface more academic and peer-reviewed material on adoption barriers specifically, which the first query is unlikely to return since it is phrased around vendor and case-study language rather than research terminology.

Filters applied: I used the Google "Tools" feature below the search bar to set a custom time range to 2024–2026 for Query 1, and Google Scholar's built-in year filter, also set to 2024–2026, for Query 2.

3. Using AI

Once the 2–3 highest-quality reports are selected, we will use an LLM (e.g., Gemini or Claude) to conduct the initial analysis. To maintain accuracy and prevent hallucinations, the AI is constrained by strict prompt strategy.

"Act as a supply-chain and AI-adoption analyst. Read the provided report [insert text/PDF] regarding AI-based demand forecasting for SMEs. Analyze the text strictly based on the following three criteria:

1. Identify the specific claims, tools, or approaches to AI-based demand forecasting described for companies of comparable size to ours.
2. List the primary risks or failure points the report highlights for SME adoption of AI forecasting.
3. Identify the methodology or primary data sources the authors used to build their conclusions.

We asked different AI to write as an example for specific prompt.